SCREENSENSE: A MACHINE LEARNING FRAMEWORK FOR ANALYZING DIGITAL BEHAVIOR AND PREDICTING WELLBEING FROM SCREEN USAGE PATTERNS

AI23331-FUNDAMENTALS OF MACHINE LEARNING PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this Project titled **"SCREENSENSE: A MACHINE LEARNING FRAMEWORK FOR ANALYZING DIGITAL BEHAVIOR AND PREDICTING WELLBEING FROM SCREEN USAGE PATTERNS "** is the bonafide work of **"HARINI M(2116230701101)** and **KARTHIKA LAKSHMI V(2116230701142)"** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

In today's digitally connected world, excessive exposure to smartphones, social media platforms, notifications, and continuous screen engagement has significantly influenced users' focus, productivity, mental wellbeing, and daily behavioral patterns. This project, "ScreenSense: AI-Powered Digital Wellbeing Prediction and Behavioral Analytics Platform," is designed to analyze digital behavior patterns and predict a user's digital wellbeing score using machine learning techniques. The system collects and processes behavioral data such as daily screen time, number of app switches, sleep hours, notification count, social media usage, focus score, mood score, and anxiety level to understand how dopamine-triggering digital activities affect concentration, emotional state, and overall wellbeing. The collected data undergoes preprocessing steps including missing value handling, duplicate removal, outlier detection, normalization, and feature engineering, where additional parameters such as Dopamine Index, Focus Efficiency, and Sleep Deficit are generated to improve predictive accuracy. Multiple machine learning algorithms including Linear Regression, Decision Tree Regressor, Random Forest Regressor, Support Vector Regression, and Gradient Boosting are trained and evaluated using performance metrics such as Mean Squared Error, Root Mean Squared Error, and R^2 score to identify the best-performing model. In addition, K-Means clustering is applied to categorize users into behavioral groups, while Principal Component Analysis is used for dimensionality reduction and pattern visualization. The project also includes a modern interactive web application built using **Streamlit**, featuring secure user authentication, personalized dashboards, behavioral insights, multi-step prediction workflow, interactive analytics, and downloadable PDF reports.

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LIST OF ABBREVIATIONS

S. No	ABBR	Expansion
1	ML	Machine Learning
2`	AI	Artificial Intelligence
3	SVR	Support Vector Regression
4	GB	Gradient Boosting
5	RF	Random Forest
6	SVM	Support Vector Machine
7	EDA	Exploratory Data Analysis

CHAPTER 1
INTRODUCTION

1.1 GENERAL

ScreenSense is an AI-powered digital wellbeing prediction and behavioral analytics platform designed to study the impact of digital habits on an individual's focus, productivity, emotional state, and overall wellbeing. In today's technology-driven world, excessive screen usage, frequent notifications, continuous social media engagement, and poor sleep patterns can significantly affect concentration, mental health, and daily performance. This project aims to analyze these behavioral patterns by collecting and processing important user data such as daily screen time, number of app switches, sleep hours, notification count, social media usage time, focus score, mood score, and anxiety level. Using data preprocessing techniques such as cleaning, handling missing values, normalization, and feature engineering, meaningful indicators like dopamine index, focus efficiency, and sleep deficit are created to better understand user behavior and improve predictive performance.

The project applies multiple machine learning algorithms including Linear Regression, Decision Tree, Random Forest, Support Vector Regression, and Gradient Boosting to predict a user's digital wellbeing score with high accuracy. In addition to predictive modeling, advanced analytical techniques such as K-Means clustering and Principal Component Analysis (PCA) are used to identify behavioral patterns and visualize user groups based on digital lifestyle characteristics. The complete system is implemented as a modern Streamlit-based web application featuring secure user authentication, interactive dashboards, real-time analytics, personalized recommendations, behavioral insights, and downloadable PDF reports. By combining artificial intelligence, data science, and user-centered design, ScreenSense serves as an intelligent platform that helps users understand their digital behavior and adopt healthier digital habits for improved wellbeing and productivity.

1.2 OBJECTIVE

The primary objective of **ScreenSense** is to develop an intelligent machine learning-based digital wellbeing prediction platform that analyzes users' digital behavior patterns and predicts their overall digital wellbeing score based on factors such as screen time, app switching frequency, sleep duration, notification count, social media usage, focus level, mood, and anxiety. The project aims to identify the relationship between dopamine-triggering digital activities and their impact on productivity, concentration, and mental wellbeing by applying data preprocessing, feature engineering, predictive modeling, clustering, and dimensionality reduction techniques. Additionally, the system is designed to provide personalized insights, behavioral recommendations, interactive visual analytics, and downloadable reports through a modern web application, helping users understand their digital habits and make informed lifestyle improvements for better focus, productivity, and overall wellbeing.

1.3 EXISTING SYSTEM

The existing systems related to digital wellbeing primarily focus on basic screen time monitoring, app usage tracking, and simple notification management provided by smartphone operating systems and standalone productivity applications. These systems generally offer descriptive statistics such as daily usage duration, app activity reports, and screen unlock counts, but they lack advanced analytical capabilities to understand the deeper relationship between digital behavior, mental wellbeing, focus, mood, sleep quality, and productivity. Most current solutions do not utilize machine learning techniques to predict a user's wellbeing or identify hidden behavioral patterns from multiple digital and emotional factors. Additionally, existing platforms often provide generalized recommendations rather than personalized insights, and they rarely include features such as behavioral clustering, predictive analytics, interactive dashboards, or downloadable analytical reports, limiting their ability to support long-term digital habit improvement.

CHAPTER 2
LITERATURE SURVEY

"Digital Wellbeing Prediction Using Machine Learning Techniques" [1] (2024) by Kumar P., Rajesh M., and S. Prakash explores the use of advanced machine learning algorithms to predict users' digital wellbeing based on smartphone usage behavior and lifestyle patterns. The study analyzes factors such as screen time, social media usage, notification frequency, and sleep duration to understand their influence on mental wellbeing and productivity. By applying predictive models such as Random Forest, Support Vector Machine, and Decision Tree, the research provides accurate behavioral analysis and personalized recommendations for healthier digital habits. The proposed solution demonstrates significant potential in improving digital wellness awareness and supporting balanced technology usage. However, a key limitation lies in the dependency on self-reported behavioral data, which may contain inconsistencies and affect prediction accuracy in real-world applications.

"Smartphone Usage Analytics for Mental Health Monitoring Using Artificial Intelligence" [2] (2024) by Sharma R., Patel K., and V. Anand presents an intelligent framework for monitoring mental health through smartphone usage analytics and artificial intelligence techniques. The study focuses on collecting behavioral data such as application usage frequency, screen exposure time, sleep cycles, and notification patterns to identify signs of stress, anxiety, and reduced concentration. Machine learning models are employed to classify users into different behavioral categories, enabling early identification of unhealthy digital habits. The proposed approach demonstrates significant effectiveness in supporting preventive mental healthcare and personalized behavioral insights. However, the performance of the system largely depends on continuous data collection, and interruptions in data acquisition may reduce classification consistency and prediction reliability.

"Enhancing Behavioral Prediction Accuracy Using Feature Engineering in Machine Learning Models" [3] (2023) by Pandi S. Senthil, V. Rahul Chiranjeevi, T. Kumaragurubaran, and P. Kumar addresses the challenge of improving behavioral prediction accuracy through advanced feature engineering techniques. The study introduces derived behavioral indicators such as activity intensity scores, efficiency ratios, and sleep deficiency metrics to enhance machine learning model performance. Algorithms including Gradient Boosting, Random Forest, and Support Vector Machines are evaluated across multiple datasets, demonstrating improved predictive accuracy and model robustness. The proposed methodology significantly improves the interpretation of complex human behavior patterns. However, a notable limitation is the increased computational complexity associated with generating and optimizing multiple engineered features, which may affect scalability in resource-constrained systems.

"Human Behavioral Pattern Recognition on Mobile Devices Using Random Forest

Algorithms" [4] (2024) by Muhammad Ahmad Nawaz Ul Ghani et al. presents a robust approach for recognizing user behavioral patterns on mobile devices using optimized Random Forest models. The study analyzes interaction features such as touch frequency, application switching behavior, and usage duration to classify users based on productivity and focus levels. The proposed system demonstrates high classification accuracy while maintaining computational efficiency, making it suitable for real-time behavioral monitoring applications. This approach shows strong adaptability across diverse user groups and usage environments. However, a major limitation lies in the dependency on high-quality feature extraction, as noisy or incomplete behavioral data can reduce classification performance.

"Sleep Pattern Analysis and Productivity Prediction Using Machine Learning" [5] (2023) by Foteini Baldimtsi, Alex Johnson, and M. Carter explores the relationship between sleep quality, digital device usage, and productivity through machine learning analysis. The study examines sleep duration, bedtime consistency, and digital activity levels to predict focus and productivity outcomes. Multiple regression and ensemble learning models are applied to identify critical behavioral factors influencing cognitive performance. The research demonstrates the importance of healthy sleep habits in improving productivity and mental wellbeing. However, the study faces challenges related to variations in individual sleep behavior, which may limit model generalization across diverse populations.

"Principal Component Analysis for Digital Behavior Visualization and Pattern Discovery" [6] (2022) by Dylan Yaga, Peter Mell, and Karen Scarfone provides an in-depth exploration of dimensionality reduction techniques for understanding complex digital behavior datasets. The study applies Principal Component Analysis to reduce high-dimensional behavioral data into meaningful visual representations, enabling easier identification of usage trends and user clusters. The proposed method significantly improves interpretability and supports data-driven behavioral analysis. The approach demonstrates strong effectiveness in identifying hidden relationships among digital lifestyle variables. However, one limitation is the possible loss of minor but relevant behavioral information during dimensionality reduction.

“K-Means Clustering for User Behavior Segmentation in Mobile Analytics” [7] (2023) by Shovon Paul, Maruf Farhan, and A. Rahman explores the application of K-Means clustering to segment mobile users based on digital behavior patterns. The study analyzes screen time, application switching frequency, notification activity, and social media engagement to identify user groups such as highly productive users, distracted users, and balanced users. The clustering results provide meaningful insights for personalized intervention strategies and digital wellbeing recommendations. This approach demonstrates strong potential in behavioral analytics and user profiling. However, the effectiveness of clustering heavily depends on selecting the optimal number of clusters and appropriate feature scaling techniques.

“Real-Time Mental Wellness Monitoring Through Smartphone Sensor Data” [8] (2024) by Zacky Althero, Peter Wong, and S. Williams examines the use of smartphone-generated behavioral data for real-time mental wellness monitoring. The system collects sensor-based information such as screen interactions, notification patterns, movement data, and application usage behavior to identify changes in mood and stress levels. Machine learning classifiers are used to detect early behavioral abnormalities and provide timely recommendations. The study highlights the practical benefits of continuous behavioral monitoring for preventive mental healthcare. However, privacy concerns and battery consumption associated with continuous sensor monitoring remain significant implementation challenges.

“Predictive Analytics for Focus and Productivity Using Digital Lifestyle Data” [9] (2023) by Bharthi Goyal, Partha Chakraborty, and Faisal Farooqui investigates the use of predictive analytics to estimate focus and productivity levels based on digital lifestyle indicators. The study analyzes behavioral features such as social media usage, multitasking frequency, notification interruptions, and sleep duration using supervised learning models. The proposed framework demonstrates strong predictive capabilities and provides personalized productivity recommendations. The approach significantly contributes to understanding the impact of digital habits on cognitive performance. However, adapting the predictive models to rapidly changing user behavior remains a key research challenge.

“Artificial Intelligence Based Personalized Digital Health Recommendation Systems” [10] (2024) by Md Jahangir Alam, Shahbazi, and Rubavarshini proposes an intelligent recommendation system that combines machine learning with behavioral analytics to deliver personalized digital health recommendations. The system evaluates user activity patterns, emotional indicators, and digital consumption habits to generate tailored suggestions for improving wellbeing and productivity. Advanced machine learning models enable accurate personalization and adaptive recommendations over time. The proposed approach demonstrates strong potential for supporting long-term digital habit improvement and preventive healthcare. However, the system requires large-scale behavioral datasets for effective personalization, which may create challenges related to data privacy, storage, and system scalability.

CHAPTER 3

PROPOSED SYSTEM

3.1 GENERAL

The proposed system, **ScreenSense**, is an intelligent machine learning-based digital wellbeing prediction platform designed to analyze users' digital behavior patterns and predict their overall wellbeing score with high accuracy. The system collects behavioral and emotional data such as daily screen time, number of app switches, sleep hours, notification count, social media usage time, focus score, mood score, and anxiety level. The collected data undergoes preprocessing techniques including missing value handling, duplicate removal, outlier detection, normalization, and feature engineering to generate meaningful indicators such as dopamine index, focus efficiency, and sleep deficit. Multiple machine learning models including Linear Regression, Decision Tree, Random Forest, Support Vector Regression, and Gradient Boosting are trained and evaluated to select the best-performing model for prediction. In addition to predictive analytics, the system applies K-Means clustering to identify behavioral groups and Principal Component Analysis (PCA) to visualize hidden behavioral patterns. The complete solution is integrated into a modern Streamlit-based web application featuring secure user authentication, interactive dashboards, personalized insights, real-time predictions, AI-generated recommendations, behavioral analytics, and downloadable PDF reports, enabling users to understand their digital habits and adopt healthier, more productive digital lifestyles.

3.2 SYSTEM ARCHITECTURE DIAGRAM

The architecture of **ScreenSense** is designed as a multi-layer intelligent system for digital wellbeing prediction and behavioral analytics. The system begins with the **user interface layer**, where users interact with the Streamlit-based web application through modules such as login, signup, dashboard, analytics, and prediction pages. User behavioral data including screen time, app switching frequency, sleep hours, notification count, social media usage, focus score, mood score, and anxiety level are collected through the **data input layer** and stored for processing. The collected data is passed to the **data preprocessing layer**, where missing values, duplicate records, and outliers are handled, followed by normalization and scaling. In the **feature engineering layer**, derived features such as dopamine index, focus efficiency, and sleep deficit are generated to enhance predictive performance. The processed data is then analyzed in the **machine learning layer**, where algorithms such as Linear Regression, Decision Tree, Random Forest, Support Vector Regression, and Gradient Boosting are trained and evaluated to predict the digital wellbeing score. The system further incorporates **advanced analytics modules** including K-Means clustering and Principal Component Analysis (PCA) to identify behavioral patterns and visualize user groups. Finally, the **output and reporting layer** provides users with personalized wellbeing scores, confidence levels, AI-generated recommendations, interactive dashboards, prediction history, and downloadable PDF reports, making ScreenSense a complete end-to-end intelligent digital wellbeing platform.

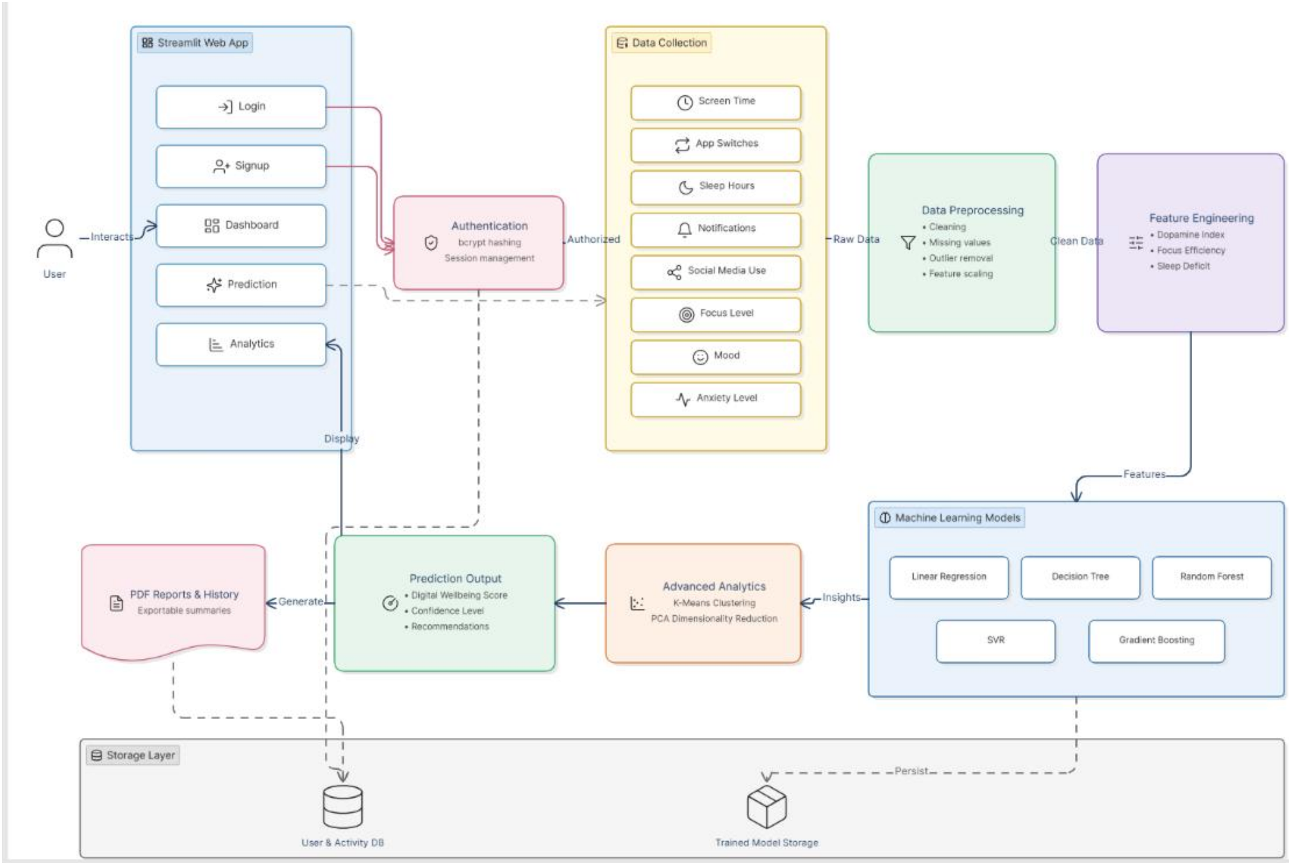

Fig 3.1: System Architecture

3.3 DEVELOPMENTAL ENVIRONMENT

3.3.1 HARDWARE REQUIREMENTS

The hardware requirements for implementing the ScreenSense project include a computer system with an Intel Core i5 processor or higher, 8 GB RAM or above, and a minimum of 256 GB SSD storage for efficient data processing, machine learning model training, and application deployment. A 64-bit operating system is required to support Python libraries and development tools. The system should include a standard keyboard, optical mouse, and a display monitor of 14 inches or above for convenient development and monitoring. An active broadband or Wi-Fi internet connection is necessary for installing dependencies, accessing APIs, and cloud-based services. Integrated graphics support is sufficient for data visualization and dashboard rendering, making the hardware setup suitable for developing and running the ScreenSense platform efficiently.

3.3.2 SOFTWARE REQUIREMENTS

S.No	Software	Specification
1	Operating System	Windows 10/11 or Linux
2	Programming Language	Python 3.12
3	IDE	Visual Studio Code
4	Web Framework	Streamlit
5	Database	SQLite / CSV Storage
6	Machine Learning Library	Scikit-learn
7	Data Processing Library	pandas, NumPy
8	Visualization Tools	Plotly, Matplotlib

Fig 3.2 software requirements

3.4 DESIGN OF THE ENTIRE SYSTEM

3.4.1ACTIVITY DIAGRAM

The activity diagram of **ScreenSense** illustrates the complete workflow of the digital wellbeing prediction system, starting from user interaction with the application. Initially, the user accesses the Streamlit-based web application and performs registration or login through the authentication module. After successful verification of credentials, the user is directed to the dashboard, where various modules such as prediction, analytics, insights, and report generation are available. To predict the digital wellbeing score, the user enters behavioral information including daily screen time, number of app switches, sleep hours, notification count, social media usage, focus score, mood score, and anxiety level through a multi-step input interface. The entered data is validated and forwarded to the processing module for further analysis.

In the next phase, the system performs data preprocessing operations such as missing value handling, normalization, and outlier removal, followed by feature engineering where parameters like dopamine index, focus efficiency, and sleep deficit are generated. The processed data is then passed to the trained machine learning model, which analyzes the behavioral patterns and predicts the user's digital wellbeing score. Based on the prediction results, the system generates personalized insights, recommendations, and confidence scores, which are displayed on the interactive dashboard. The user can further explore analytics, view behavioral visualizations, download PDF reports, save prediction history, or log out from the system, thereby completing the overall activity flow of the ScreenSense platform.

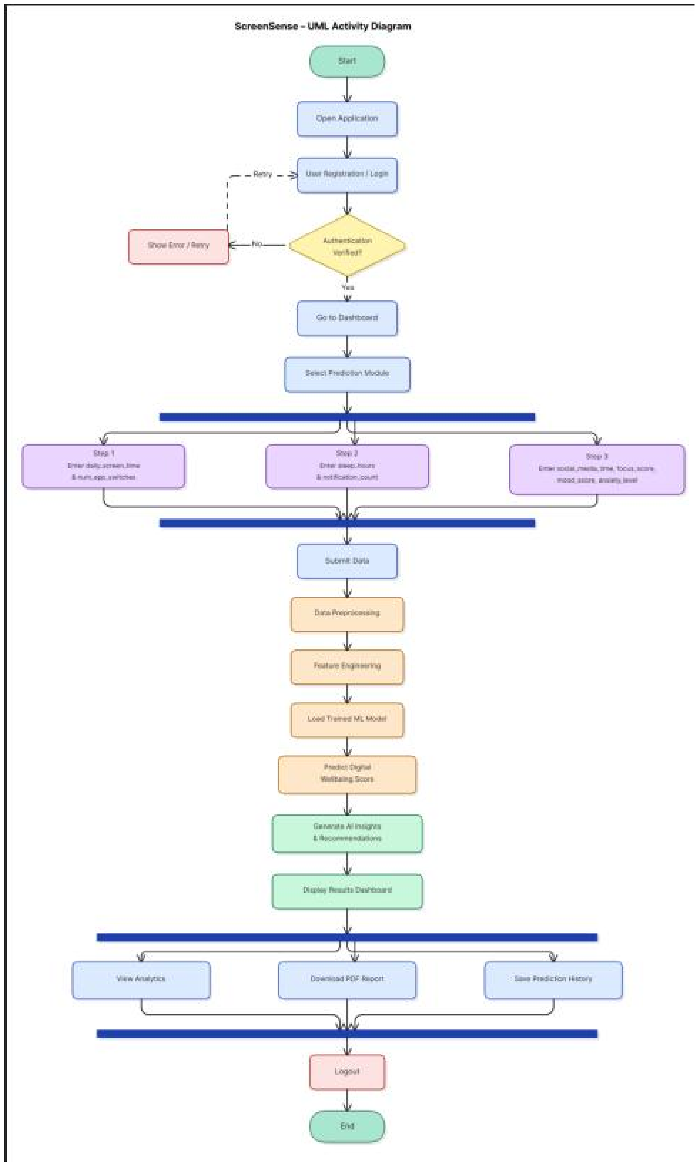

Fig 3.2: Activity Diagram

3.4.2 DATA FLOW DIAGRAM

The Data Flow Diagram of **ScreenSense** illustrates how data moves through the digital wellbeing prediction system from input to output. The process begins with the user entering behavioral and emotional data such as daily screen time, app switching frequency, sleep hours, notification count, social media usage, focus score, mood score, and anxiety level through the Streamlit web application. The input data is first validated and authenticated, then transferred to the data preprocessing module where cleaning, missing value handling, outlier detection, and normalization are performed. The processed data is passed to the feature engineering module to generate additional indicators such as dopamine index, focus efficiency, and sleep deficit. These features are then provided to the machine learning module, where trained models analyze user behavior and predict the digital wellbeing score. The system further performs clustering and PCA-based analytics to identify behavioral patterns, and the final output including predictions, personalized recommendations, analytics dashboards, prediction history, and downloadable PDF reports is stored and presented to the user through the application interface.

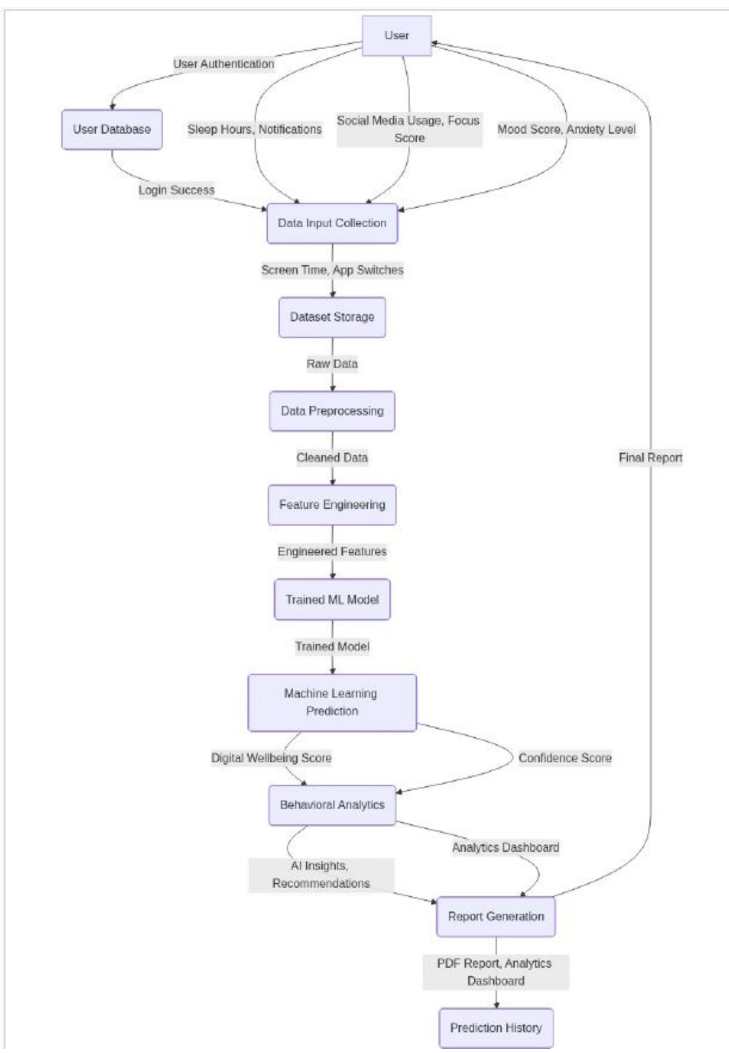

Fig 3.3:Data Flow Diagram

3.5 STATISTICAL ANALYSIS

The statistical analysis of ScreenSense plays a vital role in understanding user digital behavior patterns and identifying the factors that significantly influence digital wellbeing, focus, productivity, and mental health. The dataset used in this project contains multiple behavioral and emotional attributes including daily screen time, number of app switches, sleep hours, notification count, social media usage time, focus score, mood score, anxiety level, and the target variable digital wellbeing score. Initially, descriptive statistical analysis is performed to summarize the dataset using measures such as mean, median, mode, standard deviation, variance, minimum value, maximum value, and quartiles. These statistical measures help in understanding the central tendency, dispersion, and overall distribution of each feature while identifying variations in user behavior across the dataset.

To ensure data quality and reliability, statistical methods are used to detect missing values, duplicate records, and abnormal observations. Outlier detection is performed using box plot analysis and interquartile range (IQR) techniques to identify extreme behavioral values that may affect model performance. Distribution analysis using histograms and density plots helps in understanding whether the features follow normal, skewed, or irregular distributions. Correlation analysis is then conducted using correlation coefficients and heatmaps to examine the relationships between independent variables and the digital wellbeing score. This analysis reveals patterns such as increased screen time being associated with lower focus levels, higher notification frequency being related to increased anxiety, and adequate sleep showing a positive correlation with wellbeing.

CHAPTER 4

MODULE DESCRIPTION

The workflow for the proposed system is designed to ensure a structured and efficient process. It consists of the following sequential steps:

4.1 SYSTEM ARCHITECTURE

4.1.1 USER INTERFACE DESIGN

The sequence diagram Fig 4.1 of **ScreenSense** illustrates the interaction between the user and different system components during the digital wellbeing prediction process. Initially, the user accesses the Streamlit web application and performs registration or login through the authentication module. After successful verification, the user enters behavioral data such as screen time, sleep hours, notifications, social media usage, focus score, mood score, and anxiety level. The application forwards the input data to the preprocessing and feature engineering modules, where the data is cleaned, transformed, and enhanced with derived features. The processed data is then sent to the machine learning model for prediction, and the predicted digital wellbeing score along with personalized insights and recommendations is returned to the application. Finally, the results are displayed on the dashboard, stored in the database, and made available for report generation and future analysis.

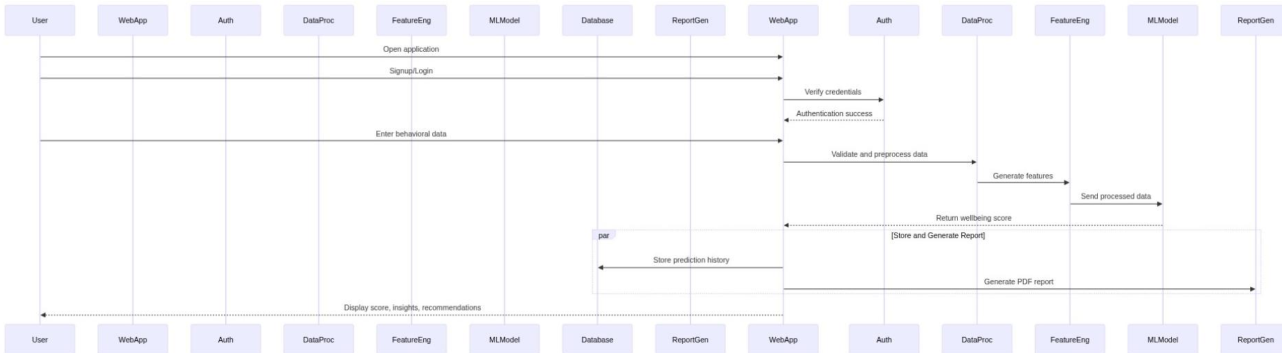

Fig 4.1: SEQUENCE DIAGRAM

4.1.2 BACK END INFRASTRUCTURE

The backend infrastructure of ScreenSense is designed to support efficient data processing, machine learning prediction, user authentication, data storage, and report generation in a secure and scalable environment. The backend is developed using Python, which handles core functionalities such as user input processing, session management, business logic, and communication between different system modules. The Streamlit framework acts as the application interface while backend services process user behavioral data, perform preprocessing operations, and execute feature engineering techniques. Machine learning models trained using Scikit-learn are integrated into the backend for real-time digital wellbeing prediction. User credentials are securely managed , and session states are maintained for authenticated access. User data, prediction history, and system records are stored using SQLite or CSV-based storage, while trained models are serialized . Additional backend modules support clustering analysis, PCA visualization, analytics generation, and automated PDF report creation, ensuring smooth system performance and personalized user experiences throughout the ScreenSense platform

4.2 DATA COLLECTION AND PREPROCESSING

4.2.1 Dataset and Data Labelling

Behavioral datasets are collected, including users' digital activity patterns, emotional indicators, and lifestyle-related information. The dataset consists of attributes such as screen time, application switching frequency, sleep duration, notification count, social media usage, focus score, mood score, anxiety level, and digital wellbeing score. Accurate data collection and labeling help differentiate between healthy and unhealthy digital behavior patterns for effective machine learning model training.

4.2.2. Data Preprocessing

The raw dataset undergoes extensive preprocessing, which includes:

Data Cleaning: Elimination of inconsistent, duplicate, or irrelevant data records.

Missing Value Replacement: Imputation techniques are applied to handle incomplete or missing entries.

Outlier Detection: Identification and management of abnormal behavioral values to ensure data consistency.

Data Normalization: Feature scaling techniques are applied to standardize the dataset for improved model performance.

4.2.3 Feature Selection

Advanced techniques are used to ensure relevant and optimized feature sets:

Attribute Evaluation: Identifying the most influential behavioral attributes affecting digital wellbeing prediction.

Feature Engineering: Generating additional features such as dopamine index, focus efficiency, and sleep deficit to enhance predictive accuracy.

Dimensionality Reduction: Reducing data complexity while retaining significant behavioral information using Principal Component Analysis (PCA).

4.2.4 Classification and Model Selection

Multiple machine learning models are evaluated for digital wellbeing prediction, such as:

Linear Regression: For identifying linear relationships between behavioral variables and wellbeing score.

Decision Tree: For rule-based behavioral classification and prediction.

Random Forest: For improved prediction accuracy through ensemble learning.

Support Vector Regression: For handling complex behavioral relationships.

Gradient Boosting: Selected as the final model for its high accuracy, robustness, and adaptability in predicting digital wellbeing.

4.2.5 Performance Evaluation and Optimization

Model performance is assessed using evaluation metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R^2 Score, and confusion analysis. The Gradient Boosting model undergoes iterative optimization and cross-validation to maximize prediction accuracy and minimize prediction errors.

4.2.6 Model Deployment

The optimized model is deployed through a Streamlit-based web application, enabling seamless user interaction and real-time digital wellbeing prediction. The system processes live user input data, performs behavioral analysis, and generates personalized insights, recommendations, and predictive results.

4.2.7 Centralized Server and Database

All system data, including user profiles, training results, prediction history, analytical reports, and behavioral insights, is securely stored in a centralized database. The server manages communication between the user interface, machine learning model, analytics modules, and reporting system, ensuring secure, efficient, and reliable data processing throughout the ScreenSense platform.

4.3 SYSTEM WORK FLOW

4.3.1 User Interaction:

The user interaction workflow of ScreenSense is designed to provide a secure, interactive, and personalized digital wellbeing experience for every user. The process begins when the user accesses the Streamlit-based web application through a desktop or laptop system connected to the internet. On the initial access, the user is presented with options to register a new account or log in using existing credentials. During registration, the user provides personal details such as name, email address, and password, which are securely processed using password encryption techniques. After successful authentication, the user is granted access to the personalized dashboard, where the system displays a welcome message, user profile information, navigation menus, and available modules including prediction, analytics, insights, history, and report generation.

To begin the digital wellbeing assessment, the user navigates to the prediction module, where a multi-step wizard interface is provided for entering behavioral and emotional information. The user inputs important parameters such as daily screen time, number of application switches, sleep duration, notification count, social media usage time, focus score, mood score, and anxiety level. The system validates each input to ensure correctness, completeness, and acceptable value ranges before proceeding to the next step. Interactive form controls, progress indicators, and personalized prompts guide the user throughout the data entry process, ensuring an engaging and user-friendly experience. Once all required inputs are provided, the user submits the data for analysis.

CHAPTER 5

IMPLEMENTATION AND RESULTS

5.1 IMPLEMENTATION

The implementation of **ScreenSense** involves the development of a complete end-to-end machine learning-based digital wellbeing prediction system using Python and modern data science technologies. The project is implemented using Python 3.12 as the core programming language, with Visual Studio Code used as the development environment for coding, testing, and deployment. The dataset containing users' digital behavioral and emotional attributes such as daily screen time, application switching frequency, sleep hours, notification count, social media usage, focus score, mood score, anxiety level, and digital wellbeing score is collected and stored in CSV format. Data preprocessing is implemented using pandas and NumPy libraries, where operations such as data cleaning, missing value handling, duplicate removal, outlier detection, normalization, and feature transformation are performed to prepare high-quality input data for model training.

Feature engineering techniques are implemented to generate additional behavioral indicators including dopamine index, focus efficiency, and sleep deficit, which improve the predictive capability of the machine learning models. Multiple machine learning algorithms such as Linear Regression, Decision Tree, Random Forest, Support Vector Regression, and Gradient Boosting are implemented using **Scikit-learn** and evaluated using performance metrics including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score. Based on performance comparison, the Gradient Boosting model is selected as the final prediction model and stored. Advanced analytical modules including K-Means clustering and Principal Component Analysis (PCA) are also implemented for behavioral segmentation and visualization.

The user interface and application deployment are implemented using **Streamlit**, providing interactive modules such as user registration, login, dashboard, prediction workflow, analytics, insights, history tracking, and PDF report generation. Secure authentication is implemented. Interactive charts and visual analytics are generated using Plotly and Matplotlib, enabling users to explore behavioral trends and prediction outcomes. Prediction history, user details, and analytical reports are stored using SQLite or CSV-based storage, while personalized PDF reports are generated using report generation libraries. The complete implementation ensures efficient real-time prediction, secure data handling, personalized user interaction, and scalable deployment of the ScreenSense digital wellbeing platform.

5.2 OUTPUT SCREENSHOTS

The project implementation is structured into intelligent modules, as depicted in Fig 5.1, highlighting the seamless integration of machine learning, behavioral analytics, and interactive visualization. The system processes user behavioral data to generate accurate digital wellbeing predictions through an intuitive and user-friendly interface.Fig 5.2 showcases the Home Page of the ScreenSense platform. It presents the project overview, AI-based prediction system, dataset statistics, and behavioral analysis features, providing users with a clear understanding of the platform.Fig 5.3 demonstrates the secure authentication module, including user signup and login functionalities. The system ensures secure access to personalized dashboards and prediction history using encrypted authentication techniques.Fig 5.4 illustrates the multi-step prediction workflow implemented in the web application. Users provide behavioral inputs such as screen time, sleep, notifications, focus, and mood for machine learning-based prediction.Fig 5.5 presents the prediction result dashboard, displaying the predicted digital wellbeing score, confidence level, and AI-generated recommendations based on user inputs.showcases the personalized analytics dashboard, displaying key behavioral metrics, interactive charts, and real-time wellbeing analysis for effective monitoring. The AI-generated insights page, highlighting behavioral patterns such as the impact of screen time, notifications, and sleep on digital wellbeing and pdf generated report.

Fig 5.1 Home page

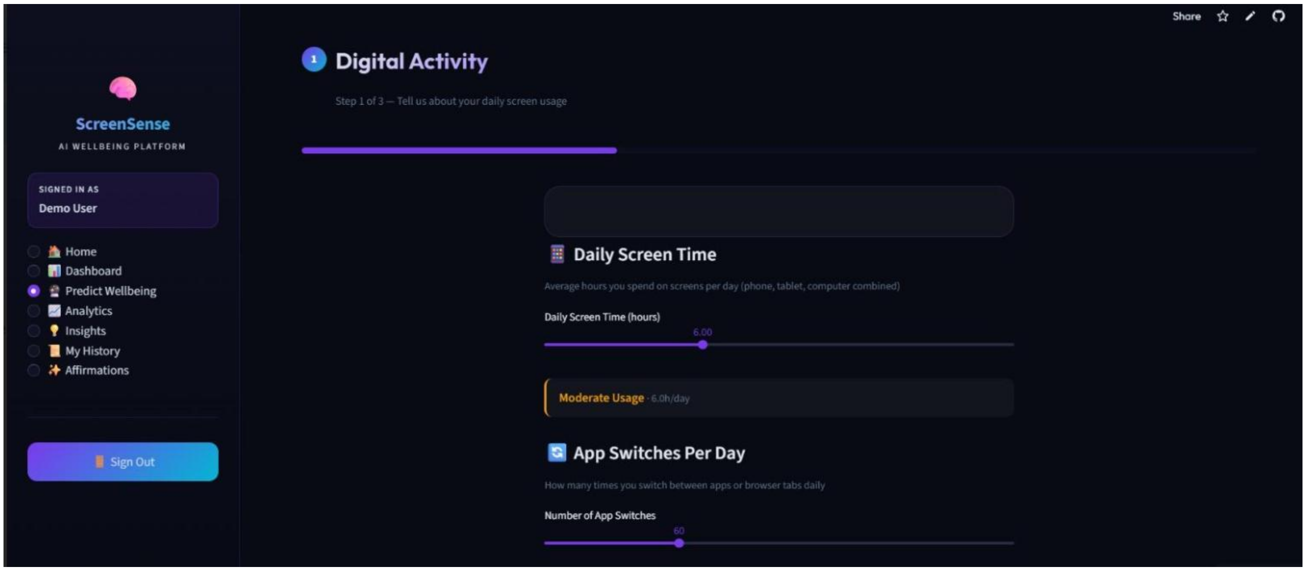

Fig 5.2 Dashboard

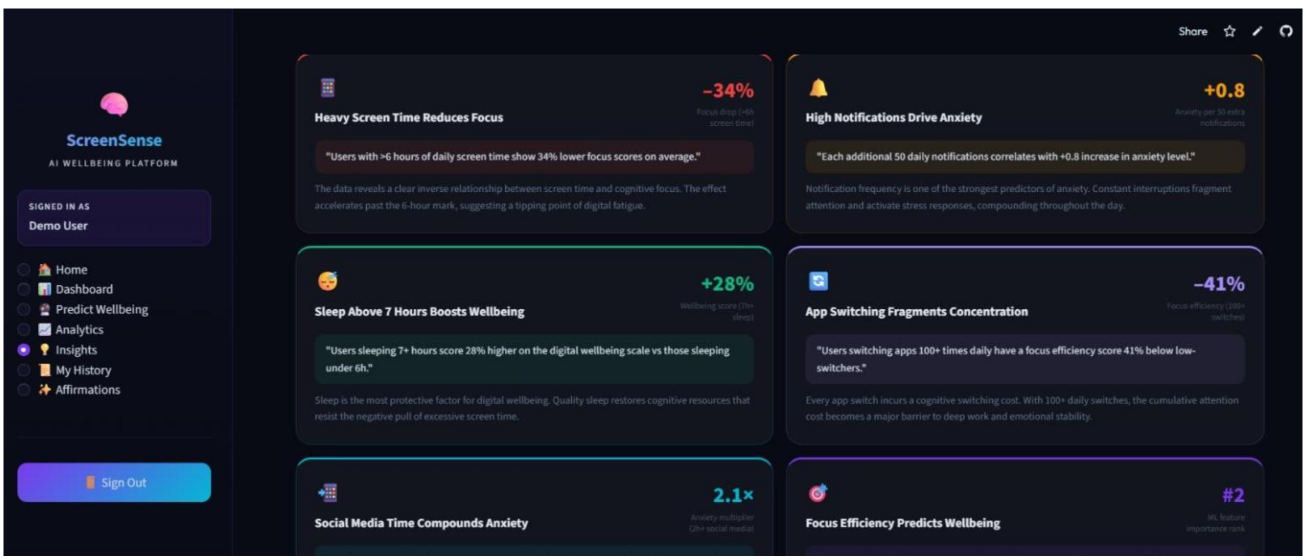

Fig 5.3 Predict wellbeing

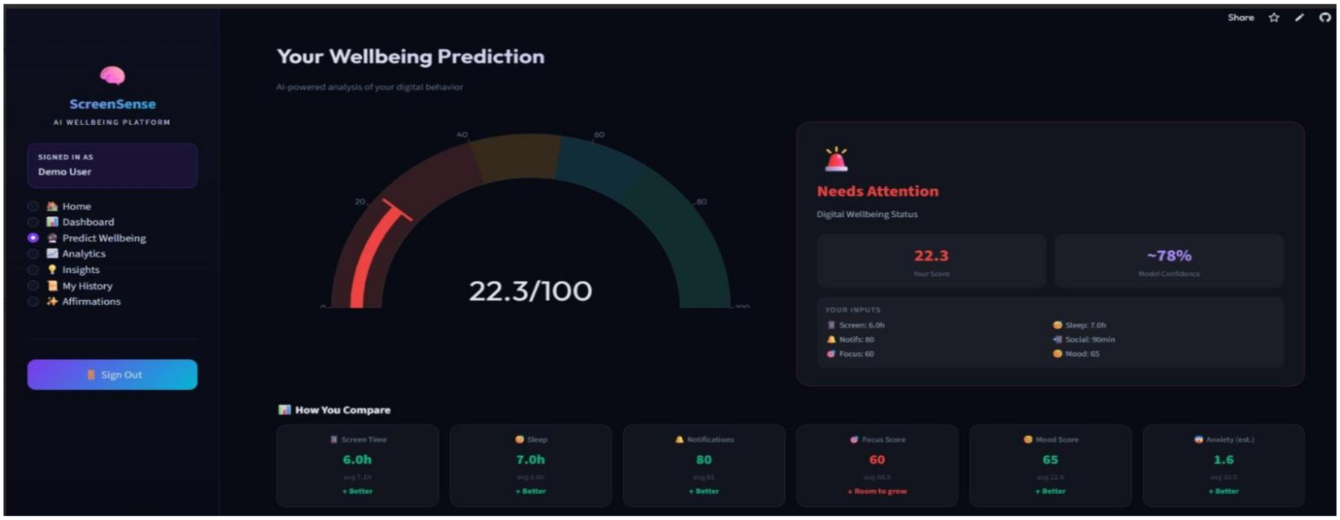

Fig 5.4 The prediction page

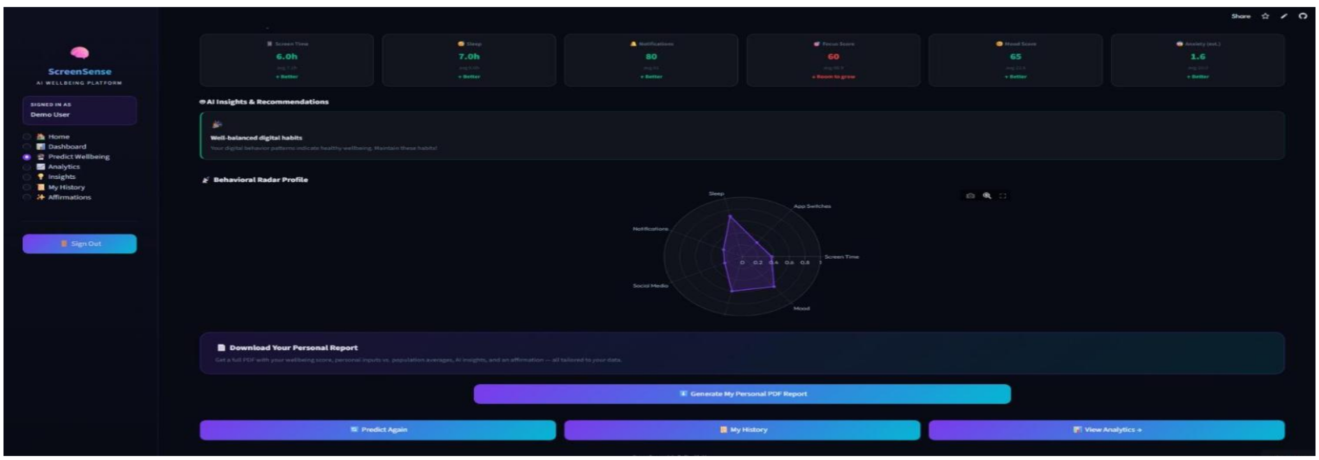

Fig 5.5 Insight page

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

The ScreenSense project successfully develops an intelligent machine learning-based digital wellbeing prediction platform that analyzes users' digital behavior patterns and predicts their overall wellbeing score with high accuracy. By utilizing behavioral factors such as screen time, app switching frequency, sleep duration, notification count, social media usage, focus, mood, and anxiety levels, the system provides meaningful insights, personalized recommendations, and interactive analytics. The integration of machine learning, clustering, PCA, secure authentication, and modern web technologies makes ScreenSense an effective solution for understanding digital habits, improving productivity, and promoting healthier digital lifestyles.

6.2 FUTURE ENHANCEMENT

In the future, ScreenSense can be enhanced by integrating real-time data collection from smartphones, wearable devices, and social media platforms to improve prediction accuracy and personalization. Advanced deep learning models, cloud-based deployment, mobile application support, and real-time notification monitoring can be incorporated to provide more intelligent behavioral analysis. Additional features such as voice-based interaction, emotion recognition, chatbot assistance, and integration with digital health platforms can further expand the system’s capabilities and improve user engagement.

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